The Mechanism of the Motion of Matter from the Bost–Connes System

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Abstract

The structure of matter and the mechanism of thermalization have been established on the state space of the Bost–Connes system; the motion of matter has so far had no definition. Starting from the hierarchy between the state space and observational spacetime, we define motion as the cycle-by-cycle displacement of the reconstruction center of a standing wave in the state space, which appears in observational spacetime as a sequence of records. Propagation proceeds in units of cycles, each cycle consisting of three steps, free evolution, formation exchange, and pattern reconstruction, and the exact dispersion relation follows; a record carries phase away through a really emitted photon, an event is the readout of a record, and time is the phase count carried by records. The conservation laws hold in the state space as operator identities and appear in observational spacetime as event-by-event balance; the representative phenomena of motion, double-slit interference, tunneling, entanglement, and macroscopic trajectories, are explained one by one from the same structure.

Contents

1 The Definition of Motion

Reference [1] starts from the state space of the Bost–Connes system and gives a unified description of spacetime, matter, and forces. The geometry of the observable space is dimension 4, signature $(1, 3)$, and the causal structure; the spectrum of the stable states gives the masses of the fermions and bosons; resonance and the metric give the couplings of the five interactions. The theory of thermalization [2] establishes, on the same state space, the mechanism of approach to equilibrium and the arrow of time; the carrier of records (the photon), its rate, and irreversibility are all theorems there. Matter in this description is a stable standing wave maintained by cycles in a thermal bath; each cycle consists of free evolution and formation exchange, and mass is the phase per cycle multiplied by the cycle rate[1].

The formation, mass, and thermalization of matter are all established in this description; the motion of matter has no definition yet. A standing wave is not a point-like object that can be displaced, and no continuous path exists between lattice sites; the old definition, an object moving along a trajectory, cannot even be stated on the discrete state space. The unresolved questions of quantum mechanics are also largely questions about motion. Electrons pass through a double slit one at a time—which slit did each pass on the way; between two observations, where is the electron; a macroscopic body has a trajectory while an electron does not—what determines the divide; what gives inertia. These questions share one source: at places where the physical process is not observed, they ask for a quantity that only observation provides. Standard quantum mechanics faces the same questions. Its Hilbert space $L^2(\mathbb{R}^3)$ is built on an externally given spacetime; the space itself has no metric and no dynamics of its own, and all structure comes from the choice of operators. Evolution proceeds continuously by the Schrödinger equation[3], while measurement is prescribed separately by projection[4]; the former yields only the wave function, the latter takes values only at the instants of observation, and the electron between two observations is described by neither. The many-body wave function is defined on the configuration space $\mathbb{R}^{3N}$, not on physical space.

This paper establishes the definition of motion on this state space.

1.1 State Space and Observational Spacetime

The state space is the arena of the motion of matter. Its structure is established in reference [1]. The lattice sites of the state space form a discrete set, and the spacing between sites is given by the logarithmic metric

$$d(m,n) = \left|\log(m/n)\right|,$$

with minimal spacing $d(1,2) = \log 2$ as the basic scale of this metric; the state space sits in a KMS thermal bath[6], whose background state is the physical vacuum; matter is a stable standing wave maintained by cycles in the bath, and propagation is carried by the convolution operator diagonal in the character basis. The metric, the fluctuations, and the dynamics are all structures of the state space itself; the formation, propagation, exchange, and thermalization of matter are each defined by these structures.

Spacetime coordinates locate the results of observation, and observation is made relative to the reference state; the reference state is the state with no mode excited, and every physical process starts from it. The structure of observational spacetime is therefore determined by the observable space at the reference state, a structure fully derived in reference [1]. The reference state is unique; its involution orbit spans the complex two-dimensional space $\mathbb{C}^2$; the observable space at the reference state is the set of all self-adjoint combinations $\mathrm{Herm}_2(\mathbb{C})$, expanded in the Pauli basis as $A = t\,I + x\,\sigma_1 + y\,\sigma_2 + z\,\sigma_3$, on which the unique invariant quadratic form is the determinant

$$\det A = t^2 - x^2 - y^2 - z^2,$$

so the dimension 4, the signature $(1,3)$, the direction of time, and the causal structure of observational spacetime are uniquely fixed; the same derivation shows at once that $\mathrm{Herm}_2(\mathbb{C})$ fixes only the geometric type, not a continuous coordinatization, and physical space remains the discrete set of lattice sites[1]. The continuity of observational spacetime has a separate origin. Its scalar field is the metric completion of the set of values, and the occupied points are countable; on the state-space side the lattice is discrete while the evolution is continuous, and both arguments are given in Appendix A. The reference state also provides the normalization for every readout; couplings and probabilities are all normalized by the background amplitude[1].

This origin settles the status of observational spacetime. The two items that generate it, the reference state and the involution, are both operator structures of the state space itself, and the derivation has no input beyond them; a geometry fully determined by the structure of another space has no independent ontological status. Observational spacetime does not carry position either: lattice positions are still carried by the state space, and a structure that carries no position cannot serve as the arena of the motion of matter. Observational spacetime is therefore not a second arena; it is the structure of readout, and what appears in it are the successive results of observation and the intervals between them.

The mathematical forms of the objects in the two spaces separate accordingly. The object in the state space is the amplitude, and the object in observational spacetime is the value obtained by observation,

$$\psi = \sum_k c_k \psi_k \ \ \text{(linear, carries phase)}\,, \qquad R = \psi\psi^\dagger \ \geq \ 0 \ \ \text{(quadratic, built from squared moduli)}\,,$$

where ψ_k is the character basis. A readout must take conjugation-invariant combinations; this is the conclusion on self-adjoint observables in reference [1]: the Galois action sends coefficients to their complex conjugates, and only combinations of type $|c_k|^2$ can be read out. The two kinds of objects belong to two levels.

That the quadratic values become probabilities is a further consequence. A single observation has no magnitude; it has only the two outcomes of occurring or not occurring, and the effective overlap takes only the values 0 and 1 ([1, Appendix A.4 (Decoherence counting)]; the two-valuedness of a record, as it is called below); probability therefore belongs not to a single observation but to the sequence of observations, as a long-time frequency. The weight of the frequency is given by equilibrium. Observation is a real exchange with the thermal bath; repeated observations refer to the same bath, and the KMS state is the unique equilibrium state of the bath ([1]; the unique stationary state in reference [2]), so the frequency of the observation sequence tends to the value of the KMS weight on the quadratic quantities. This value is exactly the normalized quadratic combination,

$$S = J\Delta^{1/2}\,, \qquad \Delta^{1/2}\cdot\Delta^{1/2} = \Delta\,, \qquad P(k) = \frac{|c_k|^2}{\sum_j |c_j|^2}\,,$$

where the polar decomposition of the modular structure splits the KMS weight into two halves, each half evaluated on amplitudes at $\beta = 1/2$, and the product of the two halves returns the full weight at $\beta = 1$; this is precisely the derivation of the Born rule in reference [1]. The normalized weight of the quadratic values equals the frequency of the observation sequence, as a consequence of the uniqueness of equilibrium, and the probability level takes its name from this.

The appearance of the two levels has a common structural source. The existence of matter presupposes the thermal bath; the equilibrium state of the bath is a KMS state, and a faithful equilibrium state necessarily carries the modular structure $S = J\Delta^{1/2}$ (the Tomita–Takesaki polar decomposition; the Born derivation of reference [1] takes it as the starting point). The antilinear factor J is complex conjugation, the operator form of the Galois restriction, so phase cannot be read out directly; the positive factor Δ carries the weights of the values. The amplitude level and the probability level are the two factors of one polar decomposition, of one origin and not juxtaposed; as long as matter exists the bath exists, the second level is delivered together with equilibrium, and its appearance depends on no choice.

The hierarchy stops at two levels. Amplitudes take values in the complex numbers, the quadratic values in the nonnegative reals; taking the squared modulus erases phase, and a further readout of nonnegative reals is again nonnegative reals, so this reduction can happen only once. The weights have only the two evaluation points $\beta = 1/2$ and $\beta = 1$, whose product returns the full weight, so the squaring closes in a single step and generates no new level. $\beta = 1$ is the unique pole of $\zeta(\beta)$, the unique phase transition[1], so the probability level has no second evaluation point; repeated action of the conditional expectation adds no shares (same appendix subsection), a readout of a value equals the value itself, and the chain of levels terminates at the second. The constraints in the reverse direction are the same. Without the amplitude level there is no interference; without the probability level there are no discrete counts; double-slit fringes and photoelectric counting require the two levels separately. The lower and upper bounds on the number of levels are therefore both two.

The relation between the two spaces is a hierarchy, not two places. The conversion between the levels is unique: from amplitude to probability the only step is the Born rule[5, 1]; the physical process that realizes the conversion is called a record, namely phase being irrevocably carried away by a really emitted photon, whose readout is called an event, with the criterion and the readings established in §3. There is no channel in the reverse direction: the quadratic combination erases phase, $|\psi|^2$ cannot recover ψ, the values contain no information to rebuild the amplitude, and the direction is one-way,

$$\text{process (amplitude level, state space)} \xrightarrow{\text{record}} \text{event (probability level, observational spacetime)}\,.$$

Observational spacetime does not carry physical processes; it only registers their records.

1.2 The Four Elements of Motion

On this two-level structure, the definition of motion consists of four elements.

- **Arena**—the arena of motion is the state space (§1.1), whose metric, lattice spacing, and fluctuations are observable structures[1].
- **Matter**—the subject of motion is a stable standing wave maintained by cycles in the thermal bath (§1.1).
- **Propagation**—the drive of motion is cyclic propagation itself: each cycle comprises one step of translation along the lattice, one exchange with the vacuum, and the reconstruction of the pattern on the lattice.
- **Record**—the record of motion is phase being irrevocably carried away by a really emitted photon, entirely within the state space; its readout is an event in observational spacetime.

The motion of matter in the state space is the displacement of the reconstruction center, driven by cyclic propagation. Propagation has two situations, and the fork lies at the exchange. When the outflowing amplitude is taken back: free evolution, formation exchange, pattern reconstruction, and the advance at the new reconstruction center; when the outflowing phase is carried away by a really emitted photon: a record occurs, the distribution of amplitude is changed, and its readout is an event in observational spacetime (§3.1). In observational spacetime, motion appears as a sequence of events.

In notation, the position distribution of matter is an amplitude on the principal-unit layer, $\psi = \sum_k c_k \psi_k$, where ψ_k is a character of the principal-unit layer and k its label (the tensor-product structure of [1]). Single-cycle propagation is the basic unit of the process of motion; each cycle consists of the three steps of free evolution, formation exchange, and pattern reconstruction[1]; the first two steps compose the transfer operator U, and the last step is the closure of the cycle, in which the acted-on components rebuild the pattern in superposition. A stable pattern is an eigenmode of U and gains one share of phase per cycle; the reconstruction center is fixed by stationary phase and moves cycle by cycle,

$$U\psi_{k,\pm} = e^{\pm i\varphi(k)}\psi_{k,\pm}\,, \qquad x_{n+1} = x_n + \frac{d\varphi}{dk}$$

($\psi_{k,\pm}$ are the two eigencombinations at label k). The full structure of motion is

$$\begin{array}{rc}
\text{state space} & \boxed{\cdots \xrightarrow{U} \psi^{(n)} \xrightarrow{\text{record } i} \tilde{\psi}^{(n)} \xrightarrow{U} \cdots \xrightarrow{U} \psi^{(n')} \xrightarrow{\text{record } i{+}1} \tilde{\psi}^{(n')} \xrightarrow{U} \cdots} \\
 & \wr\ \ \text{readout} \\
\text{observational spacetime} & \boxed{\cdots\,, \qquad R_i\,, \qquad\qquad R_{i+1}\,, \qquad \cdots}
\end{array}$$

The upper box is the entire process of motion in the state space: propagation and records alternate, a record changes the amplitude distribution ψ into $\tilde{\psi}$ (the distribution is narrowed, §5.5), and the process is not interrupted by records. The lower box is the sequence of events in observational spacetime: the readout of the i-th record is the event R_i, and records correspond to events one to one. The readout of an event comprises the impact point (position and instant) and the energy, momentum, and orientation carried by the photon; identity is not in any single event and is carried by the statistics of the event sequence.

How propagation proceeds and how records become events are the subject of the following sections.

2 Single-Cycle Propagation

The phase spectrum $\varphi(k)$ carries the entire quantitative content of motion (the structure diagram of §1.2), and $\varphi(k)$ is given by single-cycle propagation. The structure of single-cycle propagation and the kinematics it carries are the subject of this section.

2.1 The Cyclic Structure of Propagation

Propagation proceeds repeatedly in units of cycles, each cycle consisting of three steps in order,

$$(\psi^{(n)},\, x_n) \xrightarrow{\text{free evolution}} \cdot \xrightarrow{\text{formation exchange}} \cdot \xrightarrow{\text{pattern reconstruction}} (\psi^{(n+1)},\, x_{n+1})\,.$$

The standing wave is maintained by cycles in the vacuum, and propagation acts on it continually[1]; position can change only through propagation, since the modular flow σ_t changes phase only, not modulus[1], and a single excitation is not moved along the lattice by σ_t. The rigor of the cyclic structure reduces to four questions: why continual propagation has the cycle as its unit, how many sites one cycle translates, how the net transfer of the first two steps composes, and how the pattern is rebuilt at the end of the cycle. On the operator side, the dynamics is generated by the convolution K and the nearest-neighbor translation T_a, acting continually on every excitation[1].

First, the existence and exactness of the cycle unit. The exchange of a standing wave of mass m with the vacuum does not proceed continuously. The frequency-readout lemma of reference [1] shows that the band analyticity of the KMS state together with the uniqueness of lattice values forces the exchange phase to be registered cycle by cycle, exactly one share per cycle, each share exact; the share is fixed by the definition of mass,

$$\text{phase per cycle} = m/v,$$

where v is the cycle rate, shared by all excitations[1]. The argument of the lemma has two parts: the KMS condition makes correlation functions analytic in a strip of width $1/T$, so the readings of an eigenmode are restricted to functions of exponential type; a function of exponential type less than πv is uniquely determined by its values at whole cycles (Carlson-type uniqueness), so the readings proceed cycle by cycle, exactly one share per cycle, with no freedom of interpolation. Repeated application of propagation therefore has the cycle as its intrinsic unit, with period $1/v$; that the cycle is the unit is the content of the readout lemma.

Second, exactly one site per cycle. The lattice sites of position are the geometric elements of the principal-unit layer, equally spaced in the logarithmic metric (the tensor-product structure of [1]), with spacing written δ; the non-uniform spacing of §1.1 belongs to the full set of sites, and the sublattice carrying position is equally spaced. The speed of nearest-neighbor translation is fixed at $c = 1$ (metric normalization); within one cycle period $1/v$ the translated metric distance is $1/v$, which equals the lattice spacing $\delta = 1/v$ (given by the same metric normalization, [1]). At most one site per cycle is therefore an identity.

Third, the factorization of the net single-cycle transfer. Factorization comes from time ordering: within one cycle the first two steps act in order, the composition of the cycle itself fixes the ordering[1], and time-ordered evolution equals the ordered product of two exponentials; the generator $H(t)$ splits within one cycle into two segments, the first the generator of translation, the second the generator of exchange,

$$U = \mathcal{T}\exp\int_0^{1/v} H(t)\,dt = e^{i(m/v)\sigma_x}\, e^{ik\delta\sigma_z},$$

and the non-overlap of the two segments is exactly the statement that the cycle consists of the two. Free evolution acts on the position phase, formation exchange on the doublet orientation; the two generators do not commute, so the net transfer cannot merge into a single exponential. The ordered product is stable under the choice of the cycle's starting point: cutting the cycle at free evolution or at formation exchange changes only the order of the factors, the two cuts give transfer operators similar to each other ($W^{-1}UW$ type), trace and spectrum are unchanged, and the dispersion relation does not depend on the choice.

Fourth, the closure of the cycle. The mode of existence of a standing wave is reconstruction in every cycle[1]; after the first two steps have acted on each component, the components together give the pattern anew in superposition, the cycle closes by reconstruction, exactly once per cycle; how reconstruction converts phase into displacement is given, together with the transfer operator, in §2.4.

2.2 Free Evolution

Free evolution is the first step of the cycle and carries the change of position. The pattern of the standing wave moves as a whole by one site along the lattice, with form and content unchanged; free evolution contains no exchange with the vacuum, mass does not participate, and all momentum components advance at the same speed. On the operator side there are three steps.

First, the realization of translation. Free evolution is the eigenpropagation of the convolution K, preserving the form of stable states (stated in identity space in reference [1]); for an excitation carrying a position distribution, preserving form means the distribution moves as a whole, and the movement is realized by the nearest-neighbor translation T_a, one site per cycle (§2.1). Translation changes phase only and does not touch the modulus of the amplitude: $\psi_k(u+\delta) = e^{ik\delta}\psi_k(u)$ is the defining property of characters, and the phase increment $k\delta$ is exact, independent of any evaluation; the position phase does not pass through the convolution spectrum, consistent with the diagonal structure of K.

Second, direction. The two members of a doublet translate in opposite directions; the source is the relation between the involution and translation, $\iota T_a \iota = T_{-a}$, verified directly,

$$(\iota T_a \iota\, \psi)(u) = (T_a \iota\, \psi)(p-u) = (\iota\, \psi)(p-u-a) = \psi(u+a),$$

reflection turns right-movers into left-movers.

Third, the combined form. The doublet basis $\{|a\rangle, |p-a\rangle\}$ is an orbit of ι; translation acts diagonally on this basis, the two orientations gaining $e^{+ik\delta}$ and $e^{-ik\delta}$, written together as

$$\text{free-evolution factor} = e^{ik\delta\sigma_z} \qquad (\sigma \text{ are the Pauli matrices on the doublet}).$$

The free-evolution factor registers one share of translation phase on every momentum component, with opposite signs on the two branches, so one standing wave carries a left-moving and a right-moving branch at once; at zero mass the two branches do not mix and each runs straight at c—this is light. No displacement difference arises in this step; the speed differences between components come from formation exchange.

2.3 Formation Exchange

Formation exchange is the second step of the cycle, and mass enters propagation through it. In every cycle the standing wave completes one exchange with the vacuum, handing over and then retrieving the amplitude share that maintains its formation; the exchange needs no external trigger, since propagation acts on the pattern continually and modes in the same direction match in frequency and couple automatically[1]. The exchange registers one share of phase m/v per cycle on the pattern and flips the orientation of the doublet; reversal, rest, and inertia all issue from this flip (§2.5). The exchange must be compatible with the symmetry of the vacuum, and the requirement of compatibility fixes the form of the exchange completely; on the operator side there are three steps.

First, the channel. The background state ω is invariant under ι, $\omega(b) = \omega(p - b)$, so the overlap of the odd combination is

$$\langle \omega,\ \delta_b - \delta_{p-b} \rangle = \omega(b) - \omega(p - b) = 0,$$

and exchange can proceed only through even combinations—the even-entrance rule[1, Appendix A.2 (Conservation sectors of projection)]. Even combinations do not distinguish the two orientations of the doublet $\{a,\, p - a\}$, so the exchange mixes them, and the phase m/v per cycle is registered by orientation flips.

Second, the algebra. The doublet operators preserving the even channel are the image of the projection $E_{\text{even}} = \frac{1}{2}(A + \iota A \iota)$, namely $\text{span}\{\mathbf{1}, \sigma_x\}$, so the generator of the exchange has the form $\alpha \cdot \mathbf{1} + \gamma\, \sigma_x$.

Third, the balance. The modular conjugation J interchanges conjugate sector pairs and is antilinear ($S = J\Delta^{1/2}$); the background is invariant under J, so the phase inscriptions of the particle and antiparticle branches must cancel, the even channel registering $+m/v$ and the odd channel $-m/v$, forcing the trace component $\alpha = 0$; the quantitative exclusion of traceful alternatives is given in Appendix B. The generator is thereby unique,

$$\text{formation-exchange factor} = e^{i(m/v)\sigma_x}.$$

The formation-exchange factor flips orientation and registers phase; excitations that fail the balance decay away, what remains is exactly the excitations satisfying it, and detailed balance is the mode of existence of standing waves[1].

2.4 Pattern Reconstruction

Pattern reconstruction is the third step of the cycle; the cycle closes here, and one step of motion is completed with it. The momentum components acted on by the first two steps superpose on the lattice, adding where phases agree and canceling where they conflict, and the pattern is rebuilt where they add; the mode of existence of a standing wave is reconstruction in every cycle[1]. For a single eigenmode reconstruction is mere restoration; the nontrivial content appears for multi-component wave packets, motion is carried by the phase differences between components, and reconstruction converts phase into displacement.

Reconstruction is constrained by linearity. The transfer is a linear operator[1]; the components act independently and combine in superposition, and reconstruction introduces no coupling between components. The quantitative content of reconstruction is given in three steps.

First, the composition of the transfer. The ordered product of the first two steps gives the single-cycle transfer

$$U(k) = e^{i(m/v)\sigma_x}\, e^{ik\delta\sigma_z} \in SU(2)$$

(both generators are traceless, $\det e^{i\theta\sigma} = 1$, hence $U \in SU(2)$). The eigenphases $\pm\varphi(k)$ are given by the trace,

$$2\cos\varphi(k) = \operatorname{tr} U = \operatorname{tr}\big[(\cos\tfrac{m}{v} + i\sin\tfrac{m}{v}\,\sigma_x)(\cos k\delta + i\sin k\delta\,\sigma_z)\big] = 2\cos(k\delta)\cos\tfrac{m}{v}$$

(the cross terms involve $\operatorname{tr}\sigma_x$, $\operatorname{tr}\sigma_z$, $\operatorname{tr}(\sigma_x\sigma_z)$, all zero).

Second, the reconstruction center. With no phase gradient, the components agree in phase at the original place and the pattern is restored there; with a phase gradient, one cycle gives component k the extra phase $\varphi(k)$, and the place of phase agreement moves accordingly. For a wave packet $\sum_k c_k e^{i[\varphi(k)+kx]}\psi_k$, the maximum of the modulus occurs where the phase is stationary in k; the reconstruction center is fixed by

$$\frac{d}{dk}\big[\varphi(k) + kx\big] = 0$$

and the displacement per cycle is $d\varphi/dk$—this is one step of motion realized at reconstruction.

Third, the dispersion relation. Writing $E \equiv v\varphi(k)$ and $p \equiv vk\delta$, we obtain the exact dispersion relation

$$\boxed{\cos(E/v) = \cos(p/v)\,\cos(m/v)}$$

At $k = 0$, $E = m$: rest energy is the phase per cycle. That discrete-time quantum walks tend to the Dirac equation[19] is a known result (the Feynman checkerboard model[18]); here the two factors come from the first two steps of the cycle.

2.5 The Kinematics of Propagation

The speed of a standing wave, the size of its inertia, and the pace of its own time are all carried by the shape of the dispersion relation. The small-phase expansion gives $1 - \frac{E^2}{2v^2} = \left(1 - \frac{p^2}{2v^2}\right)\left(1 - \frac{m^2}{2v^2}\right) + O(v^{-4})$, i.e.

$$E^2 = m^2 + p^2 + O(p^4/v^2),$$

so the mass-shell relation is the leading shape of the dispersion, and $E = \sqrt{m^2 + p^2} \approx m + p^2/2m$ gives the correct nonrelativistic kinetic energy. The shape of the shell carries two structural remarks. The translation generator σ_z and the flip generator σ_x anticommute, so the two shares of phase combine by squares, at leading order

$$\varphi^2 \approx (k\delta)^2 + (m/v)^2\,,$$

the Pythagorean rule being the anticommutation; and the eigenvalues of $U \in SU(2)$ are always $e^{\pm i\varphi}$, the negative-energy branch being the other cycle reading of the conjugate direction $\{\chi, \bar{\chi}\}$, introducing no filled sea of negative-energy states.

First, speed. The speed of a standing wave has two layers, the fixed speed of a single branch and the average after reversals. The translation of free evolution acts necessarily in every cycle (propagation cannot be switched off, §2.1): the amplitude runs one site at c per cycle, the single-step displacement is carried by σ_z, and the eigenvalues of the velocity operator are ±1 site per cycle, i.e. $\pm c$; formation exchange can only flip the direction, it cannot slow anything down. Differentiating the dispersion relation gives

$$\sin(E/v)\,dE = \cos(m/v)\sin(p/v)\,dp\,,$$

so in the small-phase limit the group velocity is $v_g = dE/dp = p/E < 1$, and the displacement per cycle $d\varphi/dk = v_g/v$ is exactly the metric distance traveled at the group velocity in one cycle period—the average after reversals. Rest is the reading of reversals averaging to zero displacement; the strict $k = 0$ state is the completely delocalized uniform mode, and a particle at rest at some place has no corresponding state. With the carrier k_0 fixed, the displacement per cycle is constant, the exchange still balances at the displaced reconstruction center, so a uniformly moving state is a fixed point of the cycle up to allowed translations, and Newton's first law is detailed balance in the moving frame.

Second, inertia. Taking the second derivative of $E = \sqrt{m^2 + p^2}$ gives the dispersion curvature

$$\frac{\partial^2 E}{\partial p^2} = \frac{m^2}{E^3}\,,$$

which is uniquely fixed by m: inertial mass and the phase per cycle $m = v\varphi(0)$ are the same quantity. Changing speed requires changing the phase gradient, i.e. an asymmetric injection of amplitude; force enters here, and its form is established in §5.5.

Third, time. The total phase per cycle E/v splits into the translation share and the exchange share; the translation phase per cycle corresponding to group velocity p/E is $p^2/(vE)$, leaving

$$\varphi_{\text{exch}} = \frac{E}{v} - \frac{p^2}{vE} = \frac{E^2 - p^2}{vE} = \frac{m^2}{vE} = \frac{m}{v}\cdot\frac{m}{E},$$

so the exchange phase of a moving state is m/E times its rest value, and its accumulation cycle by cycle is $m\,\Delta\tau$—this is time dilation.

2.6 The Boundaries of Propagation

Propagation has two boundaries: the upper bound on speed is given by weights, and the range of validity of the dispersion relation by the spectrum.

First, the upper bound on speed. A disturbance takes time to reach a distance; propagation cannot exceed the rate of one site per cycle. A displacement of metric length ℓ per cycle carries the KMS weight $e^{-\beta\ell}$ (β the inverse temperature). The amplitude of paths reaching metric distance D within n cycles satisfies

$$|A(D,n)| \;\le\; e^{-\beta D}\,N(n,D), \qquad N(n,D) \le 2^n = e^{n\ln 2},$$

where N counts paths, at most 2^n for nearest-neighbor walks of n steps; for $D > n\ln 2/\beta$ the decay of the weight beats the growth of the count, the amplitude is exponentially suppressed, and a finite propagation speed holds (a Lieb–Robinson-type bound[8]), with an exponential tail outside the light cone. Causality holds at lattice scale to exponential accuracy and becomes strict only under continuous readout.

Second, the boundary of the spectrum. The range of validity of the dispersion relation is given by the spectrum: the eigenphase lies in $\varphi \in [0,\pi]$, so $E = v\varphi$ reaches the spectral boundary at $E = \pi v$, the same boundary as the bound $m < \pi v$ of the frequency-readout lemma of reference [1]—the same origin, the exponential type limited by Carlson-type uniqueness, two independent derivations meeting at one place. Below the spectral boundary the deviation from $E^2 = m^2 + p^2$

is $O(p^4/v^2)$, its shape fixed by the next order of the cosine expansion; the deviation of the lattice dispersion from the Lorentz dispersion is the testable item of this section.

The cyclic propagation is now complete: three steps compose the cycle, phase gradients drive displacement, the dispersion relation gives the kinematics, and the boundaries give the tests; the records that occur within propagation are established in the next section.

3 Records

§2 established cyclic propagation. A record is the physical process in which phase is irrevocably carried away by a really emitted excitation ([1, Appendix A.4 (Decoherence counting)]); it occurs within propagation (§1.2) and is completed within the state space. An event is its readout in observational spacetime, one event per record. Propagation, cycles, and the re-establishment of the standing wave are amplitude-level processes and constitute no events; the cycle rate v and the record probability α are the universal rates of propagation and of records.

Between two records there is no event in observational spacetime; no event does not mean no effect, for the phase accumulated by amplitude-level processes constrains the statistical distribution of subsequent records (§5); the photon serving as the carrier of records is itself an excitation in the state space, and the division of levels does not fall along space.

Observational spacetime—including its dimension, signature, causality, and time itself—carries not the process but the record of the process.

3.1 The Record Criterion

In every exchange of propagation, the standing wave flows its formation-maintaining amplitude share out to the vacuum (§2.3); the outflow is either taken back in detailed balance or carried away by a really emitted photon (§1.2). When the carrying-away happens is decided by three conditions; the three state, respectively, the precondition of a record, the record itself, and the result of a record.

Condition one, storage. The distinction must be written into a quantity Q invariant under propagation, the test of invariance being commutation with all observables of the layer,

$$[Q,\, A] = 0 \qquad \text{for all observables } A \text{ of the layer}\,,$$

and the list of classicizable quantum numbers is valuation, spectral class, ι-parity, and ensemble ([1, Appendix A.4 (Decoherence counting)]). That distinctions can be written only into invariants is forced by the existence condition for state-preserving readout: a φ-preserving conditional expectation onto a subalgebra $\mathcal{A}$ exists if and only if $\mathcal{A}$ is invariant under the modular flow (the Takesaki criterion[7], [1, Appendix A.2 (Conservation sectors of projection)]),

$$\exists\, E : M \to \mathcal{A},\ \varphi \circ E = \varphi \iff \sigma_t(\mathcal{A}) = \mathcal{A}\,,$$

so quantities that change with the cycle lie outside the invariant algebra, and distinctions written there admit no φ-preserving readout.

Condition two, emission. The phase must be carried off the process by a really emitted excitation, and the carrier must satisfy three conservation structures at once,

$$\chi_{\text{carrier}} = \chi_0 \text{ (character conservation)}\,, \qquad E_{\text{carrier}} = \Delta E \text{ (exact frequency match)}\,,$$
$$m_{\text{carrier}} = 0 \text{ (escapable)}\,,$$

and the excitation satisfying all three is unique: the photon (the carrier theorem, [2]). When the phase is carried away by no excitation, the environment factor is unchanged and the joint state keeps its product form,

$$(c_1\psi_1 + c_2\psi_2) \otimes \varepsilon \xrightarrow{\text{no emission}} (c_1\psi_1' + c_2\psi_2') \otimes \varepsilon\,,$$

so tracing out the environment leaves the reduced state the original superposition, and the operation equals the identity (nowhere to hand over is the identity operation). The single-record probability is the electromagnetic coupling α, so the rate at which records occur is a measurable quantity fixed by the theory.

Condition three, irreversibility. The photon does not return. The environment is renewed continually, incoming background is uncorrelated with the process, and outgoing excitations take no part in later evolution (the unique time-asymmetric assumption, [2]); the environment final states of different record histories are orthogonal share by share,

$$\langle\varepsilon'|\varepsilon\rangle = 0 \text{ (record completed)}\,, \qquad \langle\varepsilon'|\varepsilon\rangle = 1 \text{ (no record)}\,,$$

with the strict zero proved in Appendix C. When the outgoing photon is taken back, the environment factors of the two histories coincide, the overlap returns from 0 to 1, and the cross term is restored,

$$P = |A_1|^2 + |A_2|^2 + 2\,\mathrm{Re}\big[A_1\bar{A}_2\,\langle\varepsilon_2|\varepsilon_1\rangle\big]\,,$$

interference is complete at overlap 1 and the record is withdrawn; the quantum-eraser experiment[10] is the direct test of this reverse direction.

The three conditions together constitute exactly the criterion for a record. When all three hold, the share decomposition of Appendix C gives strictly orthogonal environment final states, the effective overlap takes the value 0, the cross terms vanish, and the reduced state of the system diagonalizes[9],

$$\rho = |c_1|^2\,|\psi_1\rangle\langle\psi_1| + |c_2|^2\,|\psi_2\rangle\langle\psi_2|\,,$$

the share passes to the probability level, and the record is complete. If any one fails, no record is made: a distinction written into a variable quantity has no φ-preserving readout, an operation without emission is the identity, and an emission that can return is withdrawn—the outflow is taken back in detailed balance (the first situation of §1.2). Measurement is one realization of the three conditions: the apparatus supplies the storage, receives the photon, and blocks the return, standing on the same footing as any environment; the criterion asks only the three conditions, and no observer appears in it.

3.2 Records and Events

A record is completed within the state space (§3.1) and is indivisible, the effective overlap taking only the values 0 and 1 (the two-valuedness of §1.1); its readout is therefore a single point, whose form, range, and content are fixed in three steps.

First, the form of the readout. The content of a record is determined by the structure of readout. The carrying unit of readout is the complex two-dimensional space $\mathbb{C}^2$ spanned by a doublet (the reference-state orbit of §1.1 is one instance; any pairing is isomorphic). Vectors of $\mathbb{C}^2$ are amplitudes, not readout values; a legitimate readout is invariant under the overall phase and under Galois conjugation simultaneously (§1.1), and the polynomials on $\mathbb{C}^2$ with these invariances are generated by the real combinations of $\psi_i\bar{\psi}_j$ (the first fundamental theorem of invariant theory), which are exactly the entries of the matrix $\psi\psi^\dagger$. Every readout of a record therefore factors through the map $\psi \mapsto \psi\psi^\dagger$, taking values in the space of Hermitian matrices

$$V := \mathrm{Herm}_2(\mathbb{C}) = \{R = R^\dagger\}, \qquad \dim_{\mathbb{R}} V = 4,$$

and since squared combinations presuppose the background state ($\varphi(a^*a) \geq 0$), the readout is a probability-level object from the outset.

Second, the positivity of readings. Readings must be positive, the source being the positivity of the background state. In the Pauli parametrization of §1.1, $\operatorname{tr} R = 2t$, $\det R = t^2 - x^2 - y^2 - z^2$, and the eigenvalues of R are $t \pm \sqrt{x^2+y^2+z^2}$, so

$$R \geq 0 \iff \operatorname{tr} R \geq 0 \text{ and } \det R \geq 0,$$

the set of positive readings is exactly the closed future light cone; the causal classification is given by the sign of $\det(R - R')$, and the future direction by the positivity of the trace difference. Causal structure and the future direction are the boundary of the nonnegativity of probability.

Third, the components of a reading. The identity $\det(\psi\psi^\dagger) \equiv 0$ (a rank-1 matrix has zero determinant) shows that the readout of a single amplitude state always falls on the cone surface: lightlike readings are single shares of amplitude. For a general reading ρ, the identity

$$\det \rho = \tfrac{1}{2}\left[(\operatorname{tr}\rho)^2 - \operatorname{tr}(\rho^2)\right]$$

equates the squared length of timelike intervals with the mixedness of the readout: timelike readings are mixtures. A traceless element $A = \vec{n}\cdot\vec{\sigma}$ satisfies $\det A = -|\vec{n}|^2 \leq 0$ with equality only at $A = 0$, so timelike intervals are carried entirely by the trace difference, and the reading of time coincides with the reading of total weight (calibrated in §3.3). Identity data—character direction, doublet position, and winding—do not enter the spacetime coordinates and are registered separately as internal quantum numbers; the photon carries no internal label in a record (conservation, §3.1), what it can carry is only energy–momentum, one lightlike element of V; a reading registers when and where, not what.

The form, range, and content of readout are now fully determined. The readout of one record is one positive matrix in the closed future light cone; the trace carries time, the rank carries coherence, and identity does not enter the coordinates. The affine space with V as its difference space carries all possible readouts; this affine space is observational spacetime, and its points are events. The readout of a record in the state space is an event in observational spacetime. The two items composing V, the involution and the background state, precede channel selection, so the records of all sectors share one observable space; observational spacetime is unique because the background is unique[1].

3.3 The Readings of an Event

The readings of one event fall into three kinds: the position of the impact point, the energy–momentum and orientation carried by the photon, and the instant of the impact; each kind has its own origin and form, established in turn below.

The reading of position takes spectral values. The value of a readout belongs to the spectrum of the observable; the spectrum of the position operator is the lattice, so the impact of an event falls on exactly one site each time, and the reading is discrete. The spacing of neighboring readings is the spacing of the logarithmic metric,

$$d(n, n+1) = \log(1 + 1/n)\,,$$

non-uniform in position (§1.1): the resolution of position is fixed by the lattice structure, not chosen by the instrument. The distinction of positions is carried by the direction of the scattered photon (§3.1, condition one); a single event yields one site, and the statistics of directions restores the distribution of impact points—the fringes of §5.2 are exactly this restoration made visible.

The readings of energy and momentum are locked by the lightlike condition. The frequency of the photon equals exactly the energy difference of the exchange (§3.1, condition two), so the energy read out in an event equals exactly the level difference of the process; as a reading, the photon's energy–momentum is one lightlike element of V (§3.2), trace component the energy, traceless component the momentum,

$$\det R_\gamma = 0 \iff E^2 = p^2\,,$$

so energy and momentum are not two independent readings, the lightlike condition merging them into one; orientation takes the two values of polarization, a consequence of the two winding directions of the conjugate pair[1]. What can be carried away is only energy–momentum (the conservation structure of §3.2), and this kind of reading has no other component.

The instant is the third kind of reading. The trace difference is the readout value carried by records (§3.2); its conversion to time is not yet fixed, and the calibration is completed in four steps.

First, linearity. The trace is additive on difference vectors, $R_3 - R_1 = (R_3 - R_2) + (R_2 - R_1)$, so the deposits of successive durations add; the KMS background is stationary along the modular flow (σ_t-invariant), so the deposit of a duration depends only on the duration, not on the starting point. Writing the deposit as $f(\Delta t)$, the two facts combine into the Cauchy equation,

$$f(t_1 + t_2) = f(t_1) + f(t_2)\,, \qquad f \text{ monotone} \implies f(\Delta t) = \kappa\,\Delta t$$

(of the same type as the derivation of unitary evolution in [1]; monotonicity holds because the deposit accumulates nonnegative amounts cycle by cycle). Calibration reduces to the identity of the slope κ.

Second, the clock. Stationarity also excludes the background as a clock: the readout of a stationary background contains no t. A clock can only be a periodic structure of the physical process itself, and the established periodic structure is unique—the cycle; the rate scale of thermalization belongs to the ensemble layer and is itself unconstrained by the equilibrium structure[2], so what awaits calibration cannot serve as the source of calibration; the frequency-readout lemma (appendix of [1]) guarantees that the readings proceed cycle by cycle, one logarithm per cycle, exact.

Third, records carry the count. Emission joins the phase of the source continuously (resonance is frequency matching; K is diagonal frequency by frequency, [1]). Writing Φ for the cycle phase count, the relative phase $\Delta\Phi$ of two records of the same excitation is observable (two-quantum interference); its gauge-invariant part is the accumulated flip share, the full share $m\,\Delta t$ for a state at rest, compressed to $m\,\Delta\tau$ for a moving state by the phase split of §2.5. Ramsey interference, atomic clocks, and the definition of the SI second all count cycle phase: measuring time in experiments is measuring $\Delta\Phi$.

Fourth, universality. The composition consistency of the frequency-readout lemma holds for every decomposition,

$$(m_1 + m_2)\,\Delta\tau_c = m_1\Delta\tau_1 + m_2\Delta\tau_2\,,$$

and subtracting two different decompositions gives $\Delta\tau_1 = \Delta\tau_2 = \Delta\tau_c$: Φ/m is independent of the excitation, proper time is well defined, and belongs to the history rather than to any single clock. Combining the four steps, the time coordinate is Φ/m (up to an affine constant), the identity of κ is the mass of the excitation, and the timelike interval between two records is

$$\boxed{\Delta\tau = \Delta\Phi/m} \qquad \text{proper time} = \text{phase/mass.}$$

This result agrees with the phase split of §2.5: the exchange share of a moving state accumulates cycle by cycle to exactly $m\,\Delta\tau$, two independent derivations of the same quantity, one of the consistency checks of the paper. The statement of reference [1], mass = phase per cycle × cycle rate, is here read backwards, and de Broglie's internal clock becomes a conclusion on the readout side. Each cycle registers one tick of length $1/v$, the same for all excitations (conversion between excitations is borne by the cycle rate v); in SI units $1/v \approx 2.7 \times 10^{-27}\,\mathrm{s}$ (with $v = 246.19$ GeV of [1]), so the universal step of proper-time readings is the inverse of the electroweak scale, not the Planck time—a testable item. σ_t is never read out; it shows itself only through Φ, and time is the phase count carried by records.

The full readings of one event are carried by two elements of V. The coordinates of the impact form the first element, the trace direction carrying the instant and the traceless directions the position (the parametrization of [1]); the photon's energy–momentum and orientation form the second element, lying on the cone surface. A reading has no third component; identity does not enter it (§3.2). The lattice spacing of position and the one-per-cycle of time are the minimal steps of readings, testable items of the record layer, and the dispersion deviation of §2.6 is their dynamical counterpart.

4 The Conservation Laws of Motion

Propagation and records constitute the whole process of motion, and the process keeps a set of totals unchanged; these totals are the conserved quantities. Within propagation they belong to the standing wave itself; a record transfers shares of energy, momentum, and angular momentum between matter and the photon without touching the kind, and the totals are all unchanged. Along the way in observational spacetime there are no readings (§3), so the conservation laws have no instant-by-instant form on the observational side.

4.1 Conservation Laws in the State Space

The conserved quantities in the state space are three: energy, momentum, and character charge, each given by one identity of the propagation structure. Angular momentum is not among them: rotation is not a symmetry of the lattice but a symmetry of the readout geometry, and the state space carries only the component along the intrinsic axis (§4.2); its conservation is given in the form of reading balance in §4.3.

First, the conservation of energy is given by the eigendecomposition. The energy of a standing wave neither grows nor diminishes in propagation, and exchange with the environment only transfers shares without changing the total. An eigenmode gains only phase under the single-cycle transfer,

$$U\psi_{k,\pm} = e^{\pm i\varphi(k)}\psi_{k,\pm}\,, \qquad E = v\varphi(k)$$

($\psi_{k,\pm}$ are the two eigencombinations within the doublet block of label k, §2.4); repeated application preserves the eigendecomposition, and E is unchanged cycle by cycle. The exchange with the environment conserves share by share: the interaction commutes with the total energy,

$$[H_{\mathrm{int}},\, H_S + H_B] = 0$$

(H_S, H_B the energies of system and environment), each exchange carries an exact energy quantum[2], the energy exchanged being the level difference of the cycle phase (the frequency readout of §3.3), and cooling and heating being only the transfer of shares between system and environment[2].

The energy of the cycle and the energy of the exchange issue from one operator. The generator of the modular flow is $H = \log n$[1]; for the chosen prime p, every n factors uniquely into a power of p times a coprime part, and taking logarithms gives the operator identity

$$H \;=\; (\log p)\, N \;+\; H_u$$

(N the p-adic valuation operator, H_u the logarithm of the coprime part). The shell Hamiltonian of the thermalization theory is the first term[2], the energy quantum $\log p$ of shell crossings being its spectral gap; the energy of the cycle is the spectrum of the same operator read through the cycle unit, the frequency-readout lemma giving the phase per cycle as H/v[1], so $E = v\varphi$ is the spectral value of H. The metric is the difference of the same operator, $d(m,n) = |\log(m/n)| = |H(m) - H(n)|$ (the logarithmic metric of §1.1), so energy differences and metric distances share one scale, and the KMS weight $e^{-\beta\ell}$ (§2.6) is $e^{-\beta\,\Delta E}$. The shell energy, the cycle energy, and the metric distance issue from one H, and the conservation is one law.

Second, the conservation of momentum is given by the block structure of the transfer, whose form at each momentum label is the same, independent of position. Momentum is the label of the translation phase (§2.2); the two branches of label $\pm k$ form one doublet block, and the single-cycle transfer acts block by block without mixing different blocks (§2.4),

$$U = \bigoplus_k U(k)\,, \qquad [\,U,\ \Pi_k\,] = 0$$

(Π_k the projection onto the k-th block), so the weights of the blocks of a wave packet are unchanged cycle by cycle and the distribution of momentum magnitude is conserved; the mixing of the two branches within a block is the reversal of the exchange, the sign of momentum flips with each reversal, and the average displacement is given by the group velocity (§2.5). When the lattice spacing and the cycle rate vary slowly with position, the block form varies with position and k is no longer conserved; the form of its rate of change is established in §5.5.

Third, the conservation of character charge is given by the diagonal structure of the convolution. Propagation does not change the kind of matter; an electron remains an electron after propagation of any length. The propagation operator is diagonal in the character basis[1],

$$K\chi = L(\beta,\chi)\,\chi\,, \qquad [\,K,\ P_\chi\,] = 0$$

(L the spectrum of the convolution[1], P_χ the projection onto a character direction), so the charge is unchanged cycle by cycle under propagation; formation and exchange proceed through the vacuum with character conservation as the selection rule ([1]), and the carrier of records bears no character charge (§3.1, condition two), so records do not break this conservation.

4.2 Rotation and Angular Momentum

Rotation in observational spacetime is defined on the readout space. Readings are elements of $\mathrm{Herm}_2(\mathbb{C})$, the trace carrying the instant and the three traceless directions carrying position (§3.2); among the linear maps preserving trace and det, the connected compact subgroup preserving the positive cone is the conjugation action of $SU(2)$,

$$R \;\mapsto\; WRW^\dagger\,, \qquad W \in SU(2)\,,$$

acting as $SO(3)$ on the traceless three dimensions, of the same origin as the det quadratic form[1].

Three-dimensional directions are not carried by the lattice, whose metric is a scalar (§1.1); the realization of the cyclic structure on the doublet carries a further freedom. ι is a fixed-point-free involution, traceless with spectrum ± 1; together with the translation relation $\iota T_a \iota = T_{-a}$ (§2.2) the two generate a two-dimensional irreducible representation, and representations realizing the same relations are unitarily equivalent (Schur's lemma), so all realizations form a single conjugacy class $\{WUW^\dagger,\ W \in SU(2)\}$. The dispersion relation is given by the trace (§2.4), and the trace is an invariant of the conjugacy class, so all realizations share

$$\cos(E/v) = \cos(p/v)\cos(m/v)\,,$$

whence the isotropy of the mass shell, the light speed, and the group velocity; isotropy is the statement that physical content depends only on the invariants of the conjugacy class.

Propagation in different directions consists of different realizations of one conjugacy class, and the distinction of directions is carried by the differences of the event sequence (the affine structure of §3.2). The reading of a doublet pure state is

$$\psi\psi^\dagger = \tfrac{1}{2}\big(\mathbf{1} + \hat{n}\cdot\vec{\sigma}\big)\,, \qquad \hat{n} = \langle\psi|\vec{\sigma}|\psi\rangle\,, \quad |\hat{n}| = 1\,,$$

the totality of orientations forming a sphere in one-to-one correspondence with the directions of the traceless three dimensions; for a continually recorded standing wave the events line up along the stationary-phase line (§5.5), and the difference vectors of the sequence give the direction of propagation. Directions are continuous while the radial lattice is discrete; the duality of Appendix A reappears in this division of labor.

The background takes no part in the readout of direction. On the unit-group layer section (the spectral weight state of [1, Appendix A.1 (The evaluation background and admissible evaluations)]), the background block on each doublet is computed directly as

$$\rho_D = \tfrac{1}{2}\big(\mathbf{1} + \eta\,\sigma_x\big)\,, \qquad \eta = \frac{\sum_{\text{even}} w_k - \sum_{\text{odd}} w_k}{\sum_k w_k}\,,$$

independent of the doublet's position; the σ_z component is canceled by the ι symmetry and the σ_y component by the reality of the weights, and the vanishing of the odd components is precisely the block-level form of the even-entrance rule (§2.3); the block lies inside the even algebra span$\{\mathbf{1}, \sigma_x\}$, with the bias η the difference of even and odd spectral weights. The doublet operators readable in a single record are restricted to the same even algebra, the source being the conditional expectations of the invariant subalgebra chain (the appendix subsection cited in §3.1, condition one), with which the commutant of the block agrees; components of σ_y and σ_z type cannot be read in a single record, direction can be carried only by event sequences, and the bias of the background therefore does not enter the statistics of directions.

The conserved component of angular momentum lies along the intrinsic axis of the single-cycle transfer. The transfer is a rotation about a single axis,

$$U = e^{i(m/v)\sigma_x}\, e^{ik\delta\sigma_z} = \cos\varphi + i\sin\varphi\,(\hat{e}\cdot\vec{\sigma})\,, \qquad \hat{e}\ \propto\ \big(\sin\tfrac{m}{v}\cos k\delta,\ \sin\tfrac{m}{v}\sin k\delta,\ \cos\tfrac{m}{v}\sin k\delta\big)\,,$$

with $\cos\varphi = \cos(m/v)\cos(k\delta)$ again the dispersion relation of §2.4; $[U, \hat{e}\cdot\vec{\sigma}] = 0$, the component along the axis is unchanged cycle by cycle, of the same origin as the conservation of energy (the eigendecomposition of §4.1). The axis has two limits: at $k = 0$ it lies along the exchange direction, the polarization of a standing wave at rest detaching from the direction of motion; at $m = 0$ it lies along the direction of propagation, polarization becoming helicity. The axis turns continuously with the ratio of mass to momentum, the polarization at high speed tending to the direction of propagation; the turning about the axis is one share per cycle, $\varphi = E/v$, so the precession frequency of polarization is the energy.

Rotation about the propagation axis commutes with the transfer exactly when $m = 0$. For light the rotational symmetry is exact and helicity is conserved cycle by cycle; with mass the exchange mixes the two branches, and the reversal of §2.5 reads, in angular momentum, as the flip of the spin component.

A rotation by 2π gives $W = -\mathbf{1}$: the state changes sign while the reading $\psi\psi^\dagger$ is unchanged; the relative phase of the two branches shifts by the full angle under rotation, the half angle appearing only in the overall phase. Double-valuedness enters no reading (§3.2); the transformations of the reading layer take integer spin[11], the orientation reading is two-valued (§3.3), the two values of angular momentum along the propagation direction; the state layer is double-valued while the reading layer is single-valued, a division agreeing with the two levels of amplitude and probability.

The reading of a single doublet expands only to first order in orientation; higher angular momenta are carried by tensor products of several doublets, joining the collective–relative decomposition of composites in §5.5; the full expansion of the internal spectrum of composites lies beyond this paper.

4.3 Conservation Laws in Observational Spacetime

The conservation laws hold in observational spacetime as well, in the form of balance between event readings; along the way there are no readings, so the instant-by-instant form does not exist, while the event-by-event form is complete.

Within a single event, the form of the conservation law is the lightlike lock. The photon reading satisfies $\det R_\gamma = 0$, i.e. $E^2 = p^2$ (§3.2), so the relation between energy and momentum holds inside every event.

Between events, the balance holds law by law. The process between two records conserves the quantities in the state space (§4.1); the record itself is the real emission of a photon carrying away energy and momentum, its reading the second V element of the event, equal exactly to the share exchanged (§3.3); the photon's energy–momentum is given by a single reading while that of matter is read from its event sequence (§3.3), the two kinds entering one balance. Before and after a scattering the readings sum equally,

$$\sum_{\text{out}} R = \sum_{\text{in}} R\,,$$

the photon's share and matter's share on the same ledger line.

First, the balance of energy. The trace component of the energy–momentum reading is E, and its sums before and after a scattering are equal,

$$\sum_{\text{out}} E \;=\; \sum_{\text{in}} E\,,$$

the photon's energy and the photoelectron's energy balancing together in the photoelectric effect.

Second, the balance of momentum. The traceless component of the same reading is the momentum, and its sums are equal,

$$\sum_{\text{out}} p \;=\; \sum_{\text{in}} p\,,$$

the photon occupying one term in the momentum balance of Compton scattering[12]—exactly this form.

Third, the balance of character charge. The charge does not enter single readings (§3.2); its balance appears as the counting of kinds: between two records the kind is unchanged (§4.1), the counts by kind before and after a reaction are equal, and the statistics of the event sequence carries it (§1.2).

Fourth, the balance of angular momentum, composed of two parts. The orbital part is a geometric consequence: the reading of a record is a single point (§3.2), the stationary-phase lines of free segments are straight (§2.5), and for any chosen reference event the combination of impact differences with momentum is constant along each line and joins at the event point. The orientation part issues from the destination of phase: a record is phase irrevocably carried away by the photon, phase has nowhere else to go (§3.1, condition two), the rotation phase about the event axis is a component of phase, so the change of the matter-side rotation phase must be carried by the photon's orientation (§4.2). The two parts compose the event-by-event balance of total angular momentum,

$$\sum_{\text{out}}\big[(x-x_0)\times p + h\,\hat{n}\big] \;=\; \sum_{\text{in}}\big[(x-x_0)\times p + h\,\hat{n}\big]$$

(x_0 the impact of an arbitrarily chosen reference event, $h = \pm 1$ the orientation reading, $\hat{n}$ the respective propagation direction), holding event by event through the balance of momentum and the destination of phase.

The balance holds strictly for every single event. The exchanged share is exactly one per cycle, each share exact (the frequency-readout lemma of §2.1); energy is exchanged in single-frequency quanta[2], the exchange proceeds in whole quanta, and the balance admits no continuous loosening at the single-event level. The BKS theory once weakened conservation to a property of statistical averages[14]; the coincidence experiments of Bothe and Geiger and the cloud-chamber experiments of Compton and Simon tested the balance event by event, and statistical conservation was refuted by experiment[15, 16]; strict event-by-event balance is a consequence of quantized exchange.

One conservation law has two forms, the operator identity in the state space and the reading balance in observational spacetime, the latter given by the former.

5 The Phenomena of Motion

§1 defined motion, and §2 and §3 established propagation and records. This section places that structure among the phenomena of motion, from the flight of a single electron to the arc of a stone; everyday motion is composed of quantum motion.

5.1 Elementary Quantum Motion

An electron gun fires electrons one by one into vacuum; a distant detector responds shot by shot, each response giving one position and one instant, with no reading whatever between two responses; the responding positions differ from shot to shot, and the accumulated distribution broadens with the distance of flight. The motion of a single electron is the most direct realization of the definition of §1; the following derives this set of facts one by one from the structure.

In the state space, the electron is a standing wave in the χ_2 direction carrying a position wave packet ([1]), exchanging with the vacuum and rebuilding cycle by cycle, the reconstruction center moving accordingly (§2); in observational spacetime there are only successive records, with no event between two records (§3). The full reading of an electron going from a to b follows: one record at a, one record at b, and propagation in the state space in between; along the way there is no position, position appearing only in events (§3.2).

Quantum motion has no trajectory. In the state space the process is fully determinate, the amplitude distribution evolving and the reconstruction center moving cycle by cycle (§1.2); a trajectory is a concept of observational spacetime, demanding position readouts segment by segment along the way, while the outflow along the way is taken back in detailed balance (§3.1), no record forms, no event is constituted, and no event means no position. Mott's analysis of cloud-chamber tracks[17] is the historical prototype of this reading: the seemingly continuous track is a sequence of discrete ionization records, its continuity an effect of record density in the readout. Only the general statement is given here; the double slit of §5.2 is its decisive display, and §5.5 is the limit where trajectories appear as event intervals shrink.

A freely propagating wave packet broadens cycle by cycle. The packet consists of several momentum components whose group velocities differ, the difference given by the dispersion curvature; differentiating $v_g = p/E$ (§2.5),

$$\frac{\partial v_g}{\partial p} = \frac{\partial^2 E}{\partial p^2} = \frac{m^2}{E^3} \longrightarrow \frac{1}{m} \qquad (p \ll m)\,,$$

a packet of spectral width Δp has internal group-velocity spread $\Delta v_g = (m^2/E^3)\,\Delta p$, at low speed $\Delta p/m$, so an initial width Δx_0 broadens over time t to

$$\Delta x(t) \approx \Delta x_0 + \frac{\Delta p}{m}\,t\,.$$

Spreading is a consequence of multi-component stationary phase: the stationary positions of the components differ, and the concentration of the distribution decreases cycle by cycle; the electron's small mass means large dispersion curvature, hence pronounced spreading; the inverse proportion of spreading rate to mass is contrasted with the macroscopic case in §5.5.

Quantum motion has no rest either. By §2.5, the amplitude always runs one site at c and the exchange can only flip the direction, so rest is the reading of reversals averaging to zero displacement; the strict $k = 0$ state is the completely delocalized uniform mode, and an electron at rest at some place has no corresponding state. Motion belongs to the very mode of existence of the standing wave: it can be neither added nor removed.

No trajectory, no rest, and spreading are commonly grouped under quantum uncertainty. Position and momentum are Fourier duals on a finite group (the complementarity of [1, Appendix A.2 (Conservation sectors of projection)]): the position basis and the character basis are uniform superpositions of each other, a position eigenstate at a single site spreads uniformly on the momentum side, and conversely; writing G for the principal-unit layer where position lives and $|G|$ for the number of sites, the two-sided supports of any amplitude satisfy

$$|\mathrm{supp}\,\psi| \cdot |\mathrm{supp}\,\hat{\psi}| \;\geq\; |G|$$

(the support inequality of finite-group Fourier analysis, direct from the Fourier expansion and the Cauchy–Schwarz estimate), so a narrow position distribution necessarily carries a wide momentum spectrum. The uncertainty relation is the arithmetic of duality, unrelated to the disturbance of measurement; a description sharpening both at once has no corresponding state, just as trajectory and rest have no corresponding state.

Quantum motion is propagation and records themselves, described directly by §2 and §3; how the ordinariness of macroscopic motion is composed from it is the subject of §5.5.

5.2 Double-Slit Interference

An electron source fires electrons one by one; each flies toward a barrier cut with two narrow slits, passes the apparatus, and arrives at a distant screen; each electron leaves one impact point on the screen, the positions differing shot by shot, and as impacts accumulate the long-time statistics displays alternating fringes[22]. With only one slit open, the impacts accumulate into the smooth single-slit distribution and there are no fringes; the sum of the two single-slit distributions gives no fringes either, fringes appearing only with both slits open. When a detector is placed at the slits to read out the path taken, the fringes disappear and the distribution returns to the sum of the two single-slit distributions. The following derives this set of contrasts one by one from the structure.

The departure of the electron is a formation, and its flight is the elementary motion of §5.1. The reorganization in the source proceeds along transition paths through the vacuum (§2.3), yielding a standing wave in the χ_2 direction carrying a position wave packet; the packet exchanges with the vacuum and rebuilds cycle by cycle, advancing to the barrier. This segment is identical to §5.1; what is specific to the double slit begins with the apparatus.

The barrier, the slits, and the screen the electron meets are themselves matter: composites of a great number of standing waves, carrying their own environment, places where records are being completed continually (the kinematics of composites as wholes is in §5.5); no external classical boundary remains in the narrative. The barrier acts on the incident standing wave on two levels. On the coherent level, the incident electron must rebuild within the barrier region compatibly with the local detailed balance; the exchanges in the region are occupied by the composite's many modes, modes in the same direction matching in frequency and coupling automatically[1], the local balance condition varies sharply from site to site, the rebuilt amplitude is $\sum_j w_j e^{i\theta_j}$ with phases θ_j nearly random, the expected squared modulus of the sum is $\sum_j |w_j|^2$, smaller than the coherent $|\sum_j w_j|^2$ by a factor of the mode number N, and at macroscopic mode numbers the rebuilt amplitude is suppressed to negligibility. On the probability level, the amplitude entering the barrier exchanges energy with the many modes; by the thermalization mechanism of reference [2], records fall one after another and the shares pass to the probability level—this is absorption; the screen is the fullest realization of this mechanism. A slit is a narrow channel with neither occupation nor records. The barrier suppresses, the slit passes, the screen records: the entire apparatus is given by matter.

With the apparatus included, the content of propagation is fixed. The electron standing wave advances to the barrier, still exchanging with the vacuum every cycle; reconstruction superposes the components of all paths linearly (§2.4), and after the barrier's suppression the rebuilt amplitude at screen site x retains only the two ways through the two openings, $A(x) = A_1(x) + A_2(x)$. The phases of the two ways accumulate site by site from the translation phase (§2.2, the defining property of characters), $p\,l$ over a path length l. With slits at $\pm d/2$, the path lengths from screen site x are $l_\pm = \sqrt{L^2 + (x \mp d/2)^2}$ (L the screen distance), paraxially $l_- - l_+ \approx xd/L$, so the phase difference is

$$\Delta\varphi(x) = p\,\big[l_-(x) - l_+(x)\big] \approx \frac{p\,d}{L}\,x\,,$$

and the intensity oscillates from site to site,

$$|A_1 + A_2|^2 = |A_1|^2 + |A_2|^2 + 2|A_1||A_2|\cos\Delta\varphi(x)\,,$$

with fringe spacing $2\pi L/(p\,d)$, exactly the standard formula at wavelength $\lambda = 2\pi/p$; the de Broglie wavelength[20, 21] is here the readout of the translation-phase identity. The cross term $2|A_1||A_2|\cos\Delta\varphi$ is the fringes, and it is the entire difference between the double-slit distribution and the sum of the two single-slit distributions; with one slit open the amplitude retains a single way, the impacts accumulate into the smooth $|A_1|^2$, there is no cross term, and no fringes appear.

Detection at the screen is the completion of a record (§3.1): each event falls on exactly one site, the impact taking a lattice spectral value, discrete; the Born rule gives the impact probability as the squared modulus of the amplitude, a consequence in reference [1]. Fringes belong to the amplitude, carried whole by every single electron; impacts belong to records, one per event; a great number of impacts develops the amplitude distribution point by point. Events are discrete while statistics is continuous; the two delimitations of §3 are both visible here: no event does not mean no effect, and the phase difference accumulated along the way shows itself exactly in the impact statistics.

In the state space, what exchanges and rebuilds every cycle is the amplitude of one and the same standing wave, and both openings take part in every reconstruction; which slit asks for a position along the way, but along the way no record forms and no event is constituted, so observational spacetime holds no such reading (§5.1).

The disappearance of the fringes issues from the same place. Placing a detector at the slits makes propagation complete a record there; once the record is complete, the environment's final states differ along the two ways, the photon being really emitted along one and not along the other, the two final states are orthogonal share by share, and the effective overlap takes only 0 and 1 (§3.1, condition three). The cross term is multiplied by the environment overlap,

$$P(x) = |A_1|^2 + |A_2|^2 + 2\,\mathrm{Re}\big[A_1\bar{A}_2\,\langle\varepsilon_2|\varepsilon_1\rangle\big]\ ,$$

so with the record complete, $\langle\varepsilon_2|\varepsilon_1\rangle = 0$, the two ways square first and add after, the distribution returns to the sum of the two single-slit distributions, and the fringes disappear; with no detector, $\langle\varepsilon_2|\varepsilon_1\rangle = 1$, add first and square after, and the fringes remain. The coherence–decoherence rule deciding the appearance and disappearance of fringes is the same rule used in reference [1] to compute the nine fermion masses. Appearance and disappearance are decided solely by whether the record is complete and whether the photon returns (§3.1); when the photon can be retrieved the environment overlap returns to 1 and the fringes recover—the quantum-eraser experiment; in general the fringe visibility is the modulus $|\langle\varepsilon_2|\varepsilon_1\rangle|$ of the environment overlap, continuous between 0 and 1, the completed share of the record being exactly the lost share of the fringes. Heating the barrier or admitting gas near the slits increases the records, and the fringes fade accordingly; large-molecule interference experiments record this gradation point by point[30].

Wave and particle thereby find their places: the wave is the amplitude rebuilt cycle by cycle, the particle is the single event of a completed record; one and the same standing wave, each of the two levels registering one face. Duality is not a principle; it is the name of the two levels, amplitude and probability.

5.3 Tunneling

A tip and a conducting sample are separated by a vacuum gap; the electron's energy is below the barrier of the gap, yet under a bias voltage a current flows, its magnitude falling exponentially with the gap width—the current by which the scanning tunneling microscope[23] images. An α particle is bound by the nuclear barrier, its kinetic energy below the top, yet decay occurs, the lifetime varying exponentially with the width and height of the barrier, so that lifetimes across a family of nuclides span tens of orders of magnitude[25]. In both cases the particle appears on the far side of the barrier, with probability exponentially suppressed in the width of the obstruction; the following derives this exponential law from the structure.

The barrier region acts on the incident standing wave as the barrier of §5.2 does, macroscopic mode numbers suppressing reconstruction inside the region; when the region is thick enough and has no opening, site-by-site reconstruction is entirely blocked. If translation were the only channel, the amplitude on the far side would vanish and neither current nor decay should appear.

The structure of the state space has a multiplicative direction beside the additive one: translation proceeds site by site, while one multiplicative step is not site-by-site, joining directly two positions that differ by a factor of p (the multiplicative structure of [1]; the thermalization of reference [2] proceeds along the same direction). The suppression acts on reconstruction inside the region but not on the amplitude of the multiplicative step; the pattern rebuilds directly on the far side of the barrier, with not a single reconstruction along the way.

The amplitude share through this channel per cycle is suppressed with the metric width d of the region, and the probability of the event falling on the far side takes the squared modulus by the Born rule,

$$A \;\propto\; e^{-\beta d/2}\,, \qquad P = |A|^2 \;\propto\; e^{-\beta d}\,,$$

the same exponential law in the width as the WKB transmission probability[24]; the current of the scanning tunneling microscope and the lifetime of α decay read out exactly this exponential. The traversing path has not a single reconstruction along the way, no record forms, no event is constituted, and observational spacetime holds no reading of it; the question of which path was crossed is of one kind with the question of which slit (§5.2).

5.4 Entanglement

A pair of particles is produced in a single process and flies apart in opposite directions, their separation in observational spacetime growing. Each end measures on its own: the statistics of single-end results is independent of the arrangement at the other end; but when the two sequences of results are set side by side, correlation appears, of a strength violating the Bell inequality[27], with the typical measured value 2.70[29] exceeding the classical bound 2. The following derives this set of facts one by one from the structure.

The production of the pair is one formation, and one formation can yield two standing waves at once. Formation proceeds along transition paths through the vacuum (§2.3), with character conservation as the law (§4.1); when one formation yields two excitations, the product of their identities is fixed by the conservation law while the individual identities are not. The vacuum is invariant under the involution ι, the two compatible assignments of identity stand on an equal footing and enter the superposition with equal amplitude, and the joint state has the form

$$\Psi = \frac{1}{\sqrt{2}}\big(\chi_a \otimes \chi_b + \chi_{a'} \otimes \chi_{b'}\big) \otimes g_1(x_1)\, g_2(x_2)\,, \qquad \chi_a\chi_b = \chi_{a'}\chi_{b'}\,,$$

with g_1, g_2 two separated position packets. The identity part is a sum of two equal-amplitude terms, orthogonal to each other at each end ($\chi_a \neq \chi_{a'}$). Were it a single product, its single-end reduced state would be a rank-one projection; the equal-amplitude orthogonal pair gives instead the equal-weight sum of two orthogonal projections (see the formula below), rank 2—a contradiction, so indecomposability holds. Pair emission directly furnishes the entanglement source, with no separate preparation.

Separation does not change the correlation; the argument is by the tensor-product structure. The Hilbert space factorizes into the identity factor and the position factor (the tensor-product lemma, [1]); a cyclic group has exactly one subgroup of each order, so the factorization is canonical and unique, and the divide between identity and position is not an observer's choice. The correlation lives in the identity factor, the equal-amplitude pair above; the distance lives in the position factor, the separation between the reconstruction centers of g_1 and g_2. Propagation acts on the position factor, and translation commutes with any observable O on the identity factor,

$$[\, O \otimes \mathbf{1},\; \mathbf{1} \otimes T_a\,] = O \otimes T_a - O \otimes T_a = 0\,,$$

term by term in the tensor structure. Increase or decrease of separation is realized by translation, which changes no statistics of O; what happens in observational spacetime is displacement, displacement belongs to the position factor, correlation to the identity factor, the two factors are orthogonal, and displacement does not act on correlation. Nonlocality is orthogonality.

That one end cannot read the other is a direct consequence of the hierarchy. The impact statistics at this end is given by the reduced state, the partial trace over the other end,

$$\rho_1 = \mathrm{Tr}_2 |\Psi\rangle\langle\Psi| = \tfrac{1}{2}\big(|\chi_a\rangle\langle\chi_a| + |\chi_{a'}\rangle\langle\chi_{a'}|\big) \otimes |g_1|^2 ,$$

while a measurement at the other end is a record acting on the second factor, realized by a family $\{\mathbf{1} \otimes M_i\}$ with $\sum_i M_i^\dagger M_i = \mathbf{1}$; summing over outcomes and tracing,

$$\sum_i \mathrm{Tr}_2\Big[(\mathbf{1} \otimes M_i)\,|\Psi\rangle\langle\Psi|\,(\mathbf{1} \otimes M_i)^\dagger\Big] = \mathrm{Tr}_2\Big[\,|\Psi\rangle\langle\Psi|\,\big(\mathbf{1} \otimes \textstyle\sum_i M_i^\dagger M_i\big)\Big] = \rho_1 ,$$

the partial trace absorbs all operations at the other end and ρ_1 is unchanged; single-end statistics is therefore independent of whether and when the other end measures. The correlation becomes readable only after the two record sequences are set side by side, which requires bringing the records to one place, a transport bounded by the speed limit of §2.6; no-signaling is the identity of the partial trace.

The strength of the correlation has quantitative bounds. Each end takes two dichotomic readouts, A_1, A_2 and B_1, B_2 (self-adjoint, spectrum ± 1; events take spectral values, §3.2), combined as $\mathcal{B} = A_1 \otimes (B_1 + B_2) + A_2 \otimes (B_1 - B_2)$[28]; expanding directly with $A_i^2 = B_j^2 = \mathbf{1}$,

$$\mathcal{B}^2 = 4\,\mathbf{1} - [A_1, A_2] \otimes [B_1, B_2] ,$$

each commutator has norm at most 2, so $\|\mathcal{B}\| \le \sqrt{4+4} = 2\sqrt{2}$; the Born rule reads correlation as expectation (a consequence in [1]), so $|\langle\Psi|\mathcal{B}|\Psi\rangle| \le 2\sqrt{2}$. If all readouts at both ends pass to the probability level through completed records (§3.1 condition three), the expectation reduces to a classical average over values $a_i, b_j = \pm 1$, pointwise $|a_1(b_1 + b_2) + a_2(b_1 - b_2)| = 2$, and the bound falls back to 2. Both bounds come from the structure: 2 is the arithmetic of the probability level, $2\sqrt{2}$ the norm of the amplitude-level commutators; the equal-amplitude joint state Ψ attains $2\sqrt{2}$ under suitable readouts. Measured violations exceed 2 without exceeding $2\sqrt{2}$, typically 2.70[29]; the amplitude-level commutator term takes part in the correlation in reality, its size falling exactly between the two bounds.

This phenomenon is the most direct experimental evidence for the main thesis. If physical processes are placed inside observational spacetime, correlation under separation is a puzzle[26]—how do two far-separated impacts coordinate instantly; with the process in the state space the puzzle dissolves: the correlation has been written in the identity factor since formation and has never propagated in observational spacetime, where the distance is only the readout of the position factor. Observational spacetime carries not the process but its records, and entanglement gives this claim its measurable form.

5.5 Macroscopic Motion

The phenomena of the first four subsections are all at the quantum level. Everyday motion, however, is a connected line: a stone leaves the hand and falls along an arc, the furthest thing from the quantum motion of §5.1; how the line is composed of discrete records is the subject of this subsection. Two steps are needed—the macroscopic body obeys the same propagation as a whole, and the event interval shrinks until events join into a line—argued in turn.

The first step starts from composition. A macroscopic body is a composite of a great number of standing waves, the position distribution of the i-th component being an amplitude of label k_i (the notation of §1). A global translation moves every component one site at once, and the translation phases multiply,

$$\prod_i \psi_{k_i}(u_i+\delta) = e^{i(\sum_i k_i)\delta} \prod_i \psi_{k_i}(u_i)\,,$$

so the phase increment of the composite under global translation is carried by $P = \sum_i k_i$: total momentum is the sum of component momenta. The translation phase is the defining property of characters (§2.2), and the identity holds for arbitrary interactions among the components.

The collective and relative degrees of freedom then separate by tensor product. The joint space of N components decomposes by the eigenstructure of global translation,

$$L^2(G^N) \;\cong\; L^2(G)_{\mathrm{coll}} \otimes L^2(G^{N-1})_{\mathrm{rel}}\,,$$

the collective factor labeled by the total momentum P, the relative factor generated by difference coordinates $u_i - u_j$ and invariant under global translation; whatever depends on overall position belongs to the collective factor, whatever depends on internal composition to the relative factor, the divide given by translation.

The collective factor obeys the same single-cycle transfer. Free evolution translates every component one site, the collective position moves one site, with translation phase $P\delta$; formation exchange contributes an exchange share on every component, interaction phases included, and the total phase per cycle at $P = 0$ is written M/v. The mass definition of reference [1] (phase per cycle times cycle rate) applies to the composite directly: M is the composite rest mass, binding shares automatically included. The balance of §2.3 relies only on the symmetry of the vacuum; the even-entrance rule and the J neutrality involve no number of constituents, the composite flip operator is likewise traceless; the two generators are as in §2, the single-cycle transfer has the same form,

$$U(P) = e^{i(M/v)\sigma_x}\, e^{iP\delta\sigma_z}\,, \qquad E^2 = M^2 + P^2 + O(P^4/v^2)\,,$$

and the composite lies on the same mass shell as elementary states. The center of mass moves by the same structure as a single standing wave, masses add, dispersion has the same form—this is the whole content of the stone moving as a body. Larger mass simultaneously suppresses spreading: the broadening rate $\Delta P/M$ of §5.1 is inversely proportional to mass, macroscopic masses exceed the electron's by tens of orders of magnitude, and at the same initial width the spreading time grows in proportion, so a macroscopic packet stays narrow over any observation time.

The second step is records. A macroscopic body sits in an environment that completes the three conditions of §3.1 on it continually: scattered photons write the distinction of position into their own directions—storage; the photons are really emitted, carrying phase away—emission; they escape without returning—irreversibility. The conditions hold in turn, and every scattering is an event. Superpositions across positions therefore cannot persist: were the packet in two places, the two branches would scatter the environment into different final states, orthogonal share by share (§3.1 condition three), the share passes to the probability level, and the superposition becomes a probability of one place or the other; this is the same rule by which the detector destroys the fringes in §5.2—the environment is a continually acting detector. Localization balances spreading: the packet is narrowed again and again, and a narrow packet is the equilibrium of the two rates. A billiard ball has a trajectory while an electron does not; the difference lies in two disparate ratios—mass suppresses the spreading rate, and environmental records suppress the event interval.

Trajectories thus appear, and cloud-chamber tracks (§5.1) are the case testable record by record. The initial amplitude of an α source is isotropic, no direction singled out; the first ionization event falls, the amplitude components compatible with that record concentrate into a narrow cone, momentum conservation excluding large-angle components in that single step; each subsequent ionization repeats the narrowing, and events fall one after another along the same direction. The track is a sequence of events, its straightness the shape of the stationary-phase line; isotropy and straight tracks do not conflict—the former is amplitude, the latter records. The event density of a macroscopic body exceeds any resolution, events join into a line, and seeing it move from a to b is reading exactly this sequence.

The status of Newtonian mechanics follows. With no interaction the carrier is fixed and the stationary-phase line is a uniform straight line (§2.5); interaction makes the phase per cycle vary slowly with position, the dispersion acquires position dependence, $E(P, x) = v\varphi(P, x)$. The packet center and the carrier update cycle by cycle. The total phase accumulated over n cycles is $\sum_{j\leq n} \varphi(P, x_j) + P\,(x - x_n)$; stationarity in P gives the update of the center, one cycle moving it by $\partial\varphi/\partial P$; expanding about the center, the position gradient of phase makes the accumulation rate differ from place to place, and the carrier drifts by $-\partial\varphi/\partial x$ per cycle. Reading the cycle count continuously (one tick $1/v$ per cycle, §3.3), the two update laws are the transport equations

$$\frac{dx}{dt} = \frac{\partial E}{\partial P}, \qquad \frac{dP}{dt} = -\frac{\partial E}{\partial x},$$

of which Newton's second law $F = dP/dt$ is the direct form: force is the position gradient of the phase per cycle.

Gravity is the universal case among these. A region where lattice spacing and cycle rate vary slowly with position is called, in observation, a gravitational field, and stationary-phase lines bend there. This slow variation touches no identity: the position dependence enters the $E(P, x)$ of every excitation through the same factor, the transport equations have the same form for all standing waves, and identical initial conditions give identical trajectories; everything falls alike, the equivalence principle following from universality—the stone's arc at the opening of this section is thereby explained. The position dependence of the cycle rate is at once the position dependence of time calibration, so gravitational redshift is a direct consequence of §3.3; the full derivation of the coefficients belongs to the source-side structure of the arena, beyond this paper.

The reconstruction center of a continually recorded narrow packet runs along the stationary-phase line, and this line is the trajectory; it is an emergent quantity of the record layer, belonging to no single excitation in the state space. Quantum motion is the fundamental form; macroscopic motion is its readout at two disparate ratios. What calls for explanation is not the strangeness of quantum motion but the ordinariness of macroscopic motion—and that explanation is complete.

The five phenomena issue from one structure. Elementary motion gives the prototype of propagation and records; the double slit gives the contrast between amplitude-superposed fringes and their destruction by records; tunneling gives the channel of the multiplicative direction; entanglement gives the separation of tensor factors; macroscopic motion gives the limit as event intervals shrink. In every case the observable content consists of events, and every continuous image is the statistical readout of event sequences; the definition of motion has stood the test from a single electron to a stone.

6 Motion and Light

§1 defined motion, §2 and §3 established propagation and records, and §5 completed the tests among phenomena; the subjects were all massive matter, and light has no mass. The relation of light to motion is the subject of this section.

6.1 The Propagation of Light

The zero mass of the photon is a derived result of reference [1]: the breaking direction is unique, and the orthogonal combination puts all its weight in the unbroken channel; the two polarization states are a consequence of the two winding directions of the conjugate pair[1]. Substituting $m = 0$ into the structure of §2, the entire kinematics of light reads off item by item.

Of the three steps of the cycle, the formation-exchange factor $e^{i(m/v)\sigma_x}$ becomes the identity at $m = 0$, pattern reconstruction degenerates to restoration within the advancing whole, and the single-cycle transfer retains only free evolution,

$$U(k) = e^{ik\delta\sigma_z}\,, \qquad \varphi(k) = \pm k\delta\,, \qquad E = p\,,$$

the dispersion relation $\cos(E/v) = \cos(p/v)$ having the solution $E = p$ on the principal branch $E \in [0, \pi v]$, with the massive-case $O(p^4/v^2)$ terms absent. This is light.

The consequences of the constancy of the light speed follow one by one. First, fixed speed and straight running: the group velocity $dE/dp = 1$ holds for all p, and the shell has no curvature. Second, no spreading: group and phase velocities coincide, $\partial^2 E/\partial p^2 = 0$, and the packet keeps its shape. Third, two polarizations: the single-cycle transfer is diagonal in the σ_z eigenbasis, the two branches of the doublet gain $e^{\pm ik\delta}$ and are no longer mixed by exchange, each running straight at c; the two branches are the two polarization states. Fourth, no slowing and no rest: the only mechanism of slowing is the reversal of exchange (§2.5), at $m = 0$ there is no flip, so light cannot slow down; rest is the reading of reversals averaging to zero displacement (§5.1), and here reversal itself does not exist.

Calibration fails at the same point. The phase per cycle $\varphi = k\delta$ is entirely translation share, the exchange share $m^2/(vE)$ vanishing (the phase split of §2.5), so $\Delta\tau = \Delta\Phi/m$ becomes an undefined quotient: light carries no exchange phase count and has no clock of its own. The standard conclusion that lightlike intervals have zero proper time here acquires its mechanism.

6.2 The Boundary of the Definition of Motion

Light therefore sits at the boundary of the definition of motion. Checking the four elements of §1 in turn: the arena is still the state space, records still fall one by one, appearance is still a sequence of events; of the cyclic propagation that serves as the drive, only the translation step remains—no exchange with the vacuum, no cycle-by-cycle rebuilt pattern, no standing wave maintained by cycles. Light is not the motion of matter; it is propagation itself. Conversely, every massive excitation is necessarily reversed by exchange (§2.5): massive matter is light repeatedly reversed by mass; the definition of motion covers exactly all matter, its boundary falling at the single point $m = 0$.

The boundary is two-sided. From the massive side, letting $m \to 0$, the relations $\cos(E/v) = \cos(p/v)\cos(m/v)$, $v_g = p/E$ and $\varphi_{\text{exch}} = m^2/(vE)$ degenerate term by term continuously into the kinematics of light; from the side of light, once mass is nonzero, reversal appears, and the whole structure of motion unfolds from propagation through the parameter m. The definition of motion reaches its boundary at $m = 0$ with the two sides joining continuously.

The same boundary acquires a quantitative contrast in gravity. In a non-uniform region (§5.5) the two slowly varying quantities are the lattice spacing $\delta(x)$ and the cycle rate $v(x)$, and they enter the phase at different places. The translation phase is $k\delta(x)$, where and only where the spacing enters; the number of cycles per unit time is $v(x)$, and the phase accumulation rate is the product of the phase per cycle with it. The phase per cycle thus carries two position dependences,

$$\varphi(k,x) = \underbrace{k\,\delta(x)}_{\text{translation share}} + \underbrace{\frac{m^2}{v(x)\,E}}_{\text{exchange share}}, \qquad \text{accumulation rate} = v(x)\,\varphi(k,x)\,.$$

By the split of §2.5, the fraction of the translation share in the total phase is $p^2/E^2 = v_g^2$. For slow excitations $v_g^2 \ll 1$, the position dependence of the spacing is suppressed and only that of the cycle rate counts; for light $E = p$, the phase is entirely translation share and both dependences count. The bending of stationary-phase lines follows the transverse gradient of the position dependence (the transport equations of §5.5), the ratio of shares is the ratio of bendings, and the deflection of light is twice that of the slow case. The two sides of the boundary bend alike in the same non-uniformity, with different shares.

6.3 The Unique Carrier of Records

Light carries not only propagation but records. The successive photoelectrons of the photoelectric effect[13] and the successive responses of single-photon detectors are discrete events; the countability of excitations is the first of the basic facts listed in the closing of reference [1]; and the carrier of records is unique—the excitation bearing no character charge, massless, and able to escape is only the photon (§3.1; the carrier theorem, [2]).

Uniqueness is given by the conservation laws. The defining requirement of a record is not to change the identity of what is recorded (§3.1 condition two, carrier neutrality), and candidates carrying character charge are excluded by the same conservation law. Take the neutrino: it is an excitation in the χ_3 direction carrying an identity label, so emitting a neutrino necessarily carries away one share of character charge; such a process is by definition a change of identity, not its maintenance, the total character conservation being balanced by the whole reorganization. A process that can emit a neutrino is necessarily a change of identity, while an emission completing a record must not change identity: the two kinds of process are separated by the same conservation law and do not overlap. Massive neutral candidates are excluded by the exact frequency-match clause, confined directions by the escapability clause; the three sieves exclude three kinds of candidate, and the uniqueness of the photon is the intersection of the three.

First, the rate of records. With the carrier unique, the rate has a unique calibration: the single-record probability is the electromagnetic coupling α (§3.1), and the rate at which events fall in observational spacetime is fixed by one coupling constant.

Second, the production of events. Events are not only recorded by light but produced through it: every contact at everyday scales—collision, blocking, support—involves no touching microscopically, being the coupling of the cyclic exchanges of approaching composites through the electromagnetic channel (the cross-factor channel of [1]); the phase per cycle varies sharply with position, stationary-phase lines turn, and the transport equations of §5.5 give the momentum transfer; the suppression of the incident wave by the barrier of §5.2 is the same mechanism. The strong channel is confined to nuclei, the weak channel opens only in identity-changing processes, and the interactions of everyday events pass entirely through the electromagnetic channel.

6.4 The Light Structure of Observational Spacetime

The propagation and the records carried by light merge into one structure in the geometry of observational spacetime. §3.2 gave the set of legitimate records as the closed future light cone,

$R \geq 0$; the standard decomposition of positive matrices gives

$$R = \sum_i \lambda_i \, \psi_i \psi_i^\dagger \,, \qquad \lambda_i \geq 0 \,,$$

and the rank-1 elements $\psi\psi^\dagger$ satisfy $\det(\psi\psi^\dagger) \equiv 0$, exactly the lightlike readings (§3.2: the readout of a single amplitude state always falls on the cone surface). Rank-1 elements are indecomposable: if $\psi\psi^\dagger = A + B$ with $A, B \geq 0$, both supports lie in $\operatorname{span}\{\psi\}$ and each is proportional to $\psi\psi^\dagger$; the extreme rays of the cone are therefore all lightlike, every legitimate record is a nonnegative combination of lightlike readings, and timelike readings are mixtures of two linearly independent shares (the rank criterion of §3.2)—massless excitations read on the cone surface, massive matter inside the cone. The causal classification of intervals, the future direction, and timelike lengths are all carried by the way lightlike components combine.

The boundary of the definition of motion and the appearance of motion meet here. Light sits at the boundary of the definition, with no reversal and no clock of its own; yet all events are executed by it, and all readings are carried by its lightlike components. Every readout of proper time, the phase count carried by records (§3.3), is carried off the process precisely by light; that which experiences no time carries the entire readout of time, and the calibration of time and the uniqueness of the carrier join at one and the same emission. Motion proceeds in the state space and appears, through light, in observational spacetime.

7 Conclusion

Motion in the state space is a fully determinate process. Each cycle consists of three steps: free evolution moves the pattern one site as a whole, formation exchange registers one share of phase per cycle and flips the orientation, and the pattern rebuilds where phases add; the three steps are each exact, the composed transfer operator is unitary, its trace gives the dispersion relation, and mass shell, speed, inertia, and time dilation are all its consequences. Every step of the reconstruction center is fixed uniquely by stationary phase; no trajectory, no rest, and the uncertainty relation are structural properties of this determinate process.

The only channel from the process into observational spacetime is the record. The criterion of a record is the conjunction of three conditions: when phase is not carried away by a really emitted photon, the operation equals the identity and nothing is registered in observational spacetime; the readout is a positive matrix in the closed future light cone, the dimension, signature, and causal structure of observational spacetime being fixed uniquely by the form of readout; time is the phase count carried by records, proper time equal to phase divided by mass. The channel is one-way: the quadratic combination erases phase, and readings cannot recover amplitudes.

The appearance of motion is the combination of the two. The conservation laws are the quantitative form of the combination: one law is an operator identity in the state space and an event-by-event balance of readings in observational spacetime, the balance holding strictly for every event with quantized exchange as its source, tested event by event in the coincidence experiments of Bothe and Geiger. The double slit is the decisive display of the combination: the appearance and disappearance of fringes are decided solely by whether a record is complete, with quantum erasure as the reverse test. When event intervals are long there are no readings along the way; when the intervals shrink below resolution the events join into a line; the difference between quantum and classical lies in two ratios—mass suppressing the spreading rate and records suppressing the event interval; the motion we observe is the event sequence of a determinate process.

The definition of motion is confined to identity-maintaining processes. Identity-changing processes have already appeared in the sieving of the carrier: a process that can emit a neutrino is necessarily a change of identity (§6.3); decay is exactly such a process, carried by the same state space, and its mechanism and rates are the next work.

Data Availability

No data sets were generated or analyzed in this study; all results are obtained by analytic derivation.

Conflict of Interest

The author declares no conflict of interest.

A The Duality of Discrete and Continuous

The state space and observational spacetime each occupy one side of continuity. The lattice of the state space is discrete while its evolution is continuous; the carrying geometry of observational spacetime is continuous while the occupation by events is countable; discrete and continuous exchange places exactly between the two spaces.

The discreteness of the state-space lattice is constructive. Its domain is a finite set of sites, and both the position basis and the character basis are finite,

$$\dim L^2(\mathbb{Z}/p^K\mathbb{Z}) = p^K < \infty$$

([1]).

The evolution of the state space is continuous. H is self-adjoint and bounded on a finite-dimensional space, so $t \mapsto \sigma_t = e^{itH}$ is a norm-continuous one-parameter unitary group,

$$\|\sigma_{t+\epsilon} - \sigma_t\| = \|e^{i\epsilon H} - \mathbf{1}\| \;\leq\; |\epsilon|\,\|H\| \;\longrightarrow\; 0 \qquad (\epsilon \to 0)\,,$$

the amplitude coefficients vary continuously with t, and amplitudes take values in $\mathbb{C}$. Both continuous items live at the amplitude level. The readout of the stationary background contains no t (§3.3): the evolution parameter cannot be read directly, showing itself only through the phase count, which proceeds cycle by cycle, exactly one share per cycle ([1]).

A single reading in observational spacetime falls in a countable set. The readings of one event comprise the impact point, the instant, and the photon's energy–momentum (§1.2, §3.3); position and energy–momentum take spectral values, the instant takes the phase count,

$$\#\{\text{position readings}\} = p^K\,, \qquad E \in \{\log n :\; n \in \mathbb{N}^*\}\,, \qquad \tau = \Phi/m \in \tfrac{1}{m}\,\mathbb{Z}_{\geq 0}\,,$$

with the frequency-readout lemma guaranteeing the count cycle by cycle, exactly one per cycle ([1]). In a finite span the number of cycles is finite and the number of events does not exceed it; the totality of readings is at most countable.

The totality of readings is countable yet dense. For fixed K the reading set is finite, and the metric completion of a finite set is still finite, so continuity does not come from any fixed K; the source of density is the non-uniformity of the metric together with the unboundedness of K. The spacing of neighboring sites decreases monotonically to zero,

$$d(n, n+1) = \log(1 + 1/n) \;\searrow\; 0 \qquad (n \to \infty)$$

([1]), so the intervals between readings have no common minimum; the extension across K is carried by multiplication by p ([1]), and the totality of interval readings is dense on the real line,

$$\{\log(m/n) : \ m, n \in \mathbb{N}^*\} = \log \mathbb{Q}_+ \subset \mathbb{R} \text{ dense } :$$

for any real x and $\varepsilon > 0$, choosing a rational $q \in (e^x, e^{x+\varepsilon})$ gives $\log q \in (x, x + \varepsilon)$.

That observational spacetime has $\mathbb{R}$ as its scalar field is not an independent assumption. The reading set embeds densely and isometrically in $\mathbb{R}$, and its metric completion is isometric to $\mathbb{R}$, unique up to isometry,

$$\overline{\log \mathbb{Q}_+} = \mathbb{R}\,,$$

so the geometry carried by observational spacetime—dimension 4, signature $(1, 3)$, and the causal structure—is built on the closure of the reading set, with det extended continuously from the dense subset, the extension unique. This is the same mechanism as the construction of the real field itself (the completion of $\mathbb{Q}$): the continuum arises as the completion of discrete arithmetic.

The compulsion to complete comes from the statistical layer. Single readings fall in a countable set, on which both the affine structure and det are definable, so completion is not needed for single records; the numbers of statistics are Born-weighted sums over spectral values,

$$\langle x \rangle = \sum_i p_i x_i\,, \qquad p_i = \frac{|c_i|^2}{\sum_j |c_j|^2}\,,$$

expectations, variances, and fringe visibilities all take this form, their values in general not in $\log \mathbb{Q}_+$; weak limits of empirical distributions are supported on the closure, and the range of convergence statements is the completed space. Every number of the statistical layer requires the real field; without the closure these numbers in general do not exist.

The state space is continuous amplitude on a discrete lattice: domain discrete, values and evolution parameter continuous, all the continuous parts not directly readable; observational spacetime is a countable event set on a continuous carrier: carrying geometry continuous, occupation countable. Continuity lives entirely in the amplitude level and the carrying geometry, countability entirely in events and readings,

$$\begin{gathered}\text{continuous: amplitude, phase, evolution parameter, carrying geometry}\\ \text{countable: sites, events, readings, cycle counts}\,,\end{gathered}$$

a divide coinciding with the bookkeeping divide of the amplitude and probability levels. Whether observational spacetime is discrete is thereby answered by layer: the carrying geometry is continuous, the occupation by events countable; the discreteness of the substrate leaves testable items at the boundary of the readout domain—the deviation of the lattice dispersion from the Lorentz dispersion and the exponential tail outside the light cone (§2.6).

B Exclusion of Traceful Alternatives

The balance step of §2.3 requires the generator of formation exchange to be traceless; traceful alternatives are excluded on both the structural and the numerical side. The general even-channel generator has the form $\alpha \cdot \mathbf{1} + \gamma\,\sigma_x$ (§2.3, second step). Extracting the trace component as a global phase factor, the single-cycle transfer is

$$U(k) = e^{i\alpha}\, e^{i\gamma\sigma_x}\, e^{ik\delta\sigma_z}\,,$$

with eigenphases

$$\varphi_\pm = \alpha \pm \psi(k)\,, \qquad \cos\psi = \cos(k\delta)\cos\gamma\,.$$

For $\alpha \neq 0$ the spectrum is no longer paired and the particle and antiparticle branches are asymmetric; the modular conjugation J requires the phase inscriptions of the two branches to cancel (§2.3, third step), i.e. $\varphi_+ = -\varphi_-$, forcing $\alpha = 0$.

The numerical exclusion is equally clear. Take the representative traceful form, registering the whole phase per cycle on the even projection,

$$C = e^{i(m/v)P_+}\,, \qquad P_+ = \tfrac{1}{2}(\mathbf{1} + \sigma_x)\,, \qquad \alpha = \gamma = \frac{m}{2v}\,,$$

whose positive-energy dispersion is

$$E(k) = \frac{m}{2} + \sqrt{\frac{m^2}{4} + p^2} + O(p^4/v^2)\,.$$

At $k = 0$, $E = m$ agrees with the rest energy, but three failures appear at once. First, the nonrelativistic expansion gives kinetic energy p^2/m, twice the correct $p^2/2m$; second, a constant offset $m/2$ survives at large momentum and the mass shell $E^2 = m^2 + p^2$ fails; third, the negative-energy branch has $E_-(0) = 0$, the antiparticle spectrum collapsing to zero frequency. The three failures share one source, the trace component, and the exclusion is the quantitative content of the balance step.

C Orthogonality of Environment Final States

Condition three of §3.1 asserts the orthogonality of the environment final states of different record histories; this orthogonality is a structural strict zero. Let the two histories complete their records in cycles n and n' ($n \neq n'$); the outgoing photons occupy the environment's n-th and n'-th batches of shares. The environment is renewed cycle by cycle, incoming shares uncorrelated with the process (the unique time-asymmetric assumption, [2]). On the n-th batch the two final states are respectively an occupied state and the vacuum, so the single-share inner product vanishes. Writing $|e_k^{(n)}\rangle$ for the factor of history n's final state on the k-th batch, the joint final state factorizes along shares,

$$\langle \varepsilon_{n'} | \varepsilon_n \rangle = \prod_k \langle e_k^{(n')} \mid e_k^{(n)} \rangle\,, \qquad \langle e_n^{(n')} \mid e_n^{(n)} \rangle = \langle 0|1\rangle = 0\,,$$

and one vanishing factor annihilates the whole,

$$\langle \varepsilon_{n'} | \varepsilon_n \rangle = 0\,, \qquad n \neq n'\,,$$

a strict zero rather than a small quantity. That the outgoing photon does not return (the second half of the same assumption) preserves the orthogonality to the end of the process; the zero branch of the "effective overlap takes only 0 and 1" of [1, Appendix A.4 (Decoherence counting)] is thereby realized. Alternatives would require different batches of shares to coincide or photons to return, contradicting the renewal of the environment, and are excluded.

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